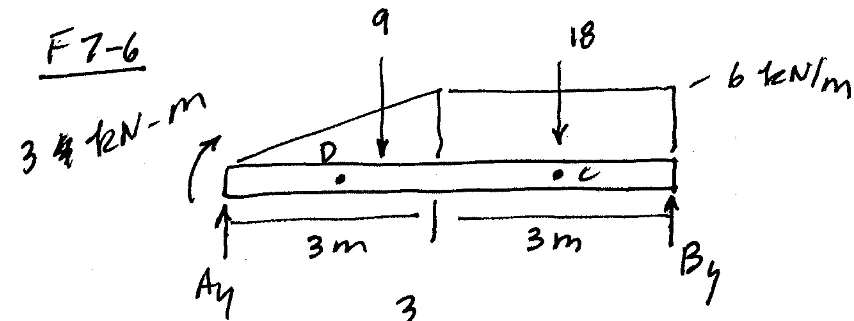
video

F7-6

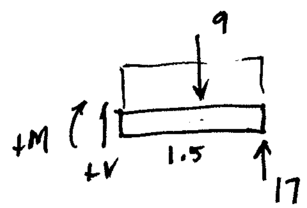


$$+\circlearrowleft \Sigma M_A = 0; \quad \cancel{4}\,3 + 9(2) + 18(4.5) - B_y(6) = 0$$

$$B_y = \frac{102}{6} = 17 \text{ kN}$$

$$A_y = 10 \text{ kN}$$

C



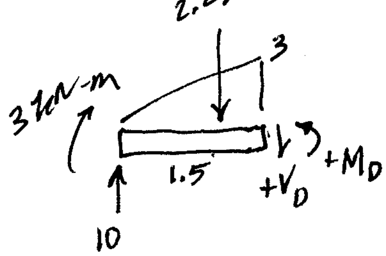
$$+\uparrow \quad V - 9 + 17 = 0$$

$$\boxed{V_C = -8 \text{ kN}}$$

$$+\circlearrowleft \Sigma M_{cut} = 0; \quad M_C + 9(.75) - 17(1.5) = 0$$

$$\boxed{M_C = 18.75 \text{ kN}\cdot\text{m}}$$

D



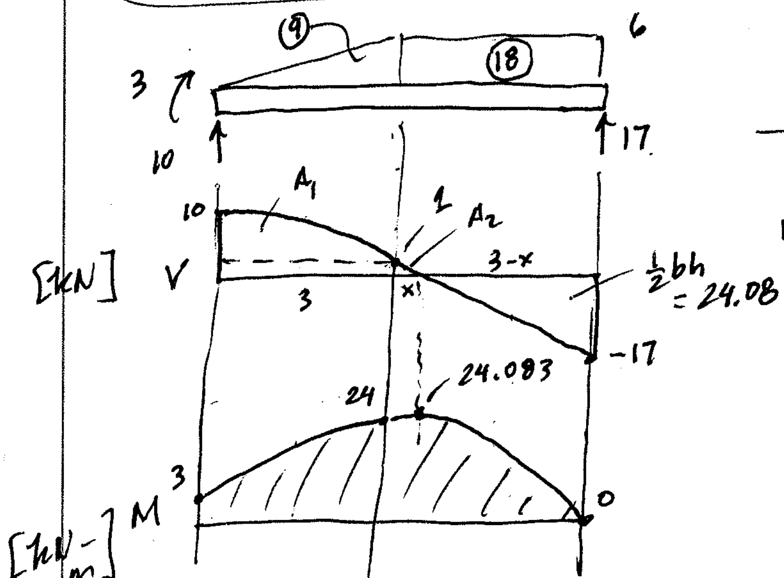
$$+\uparrow \quad 10 - 2.25 - V_D = 0$$

$$\boxed{V_D = 7.75 \text{ kN}}$$

1.125

$$+\circlearrowleft \Sigma M_{cut} = 0; \quad 3 + 10(1.5) - 2.25(.5) - M = 0$$

$$\boxed{M_D = 16.875 \text{ kN}\cdot\text{m}}$$



$$A_1 = \tfrac{2}{3}(9)(3) + 1(3) = 21$$

$$\frac{x}{1} = \frac{3-x}{17}$$

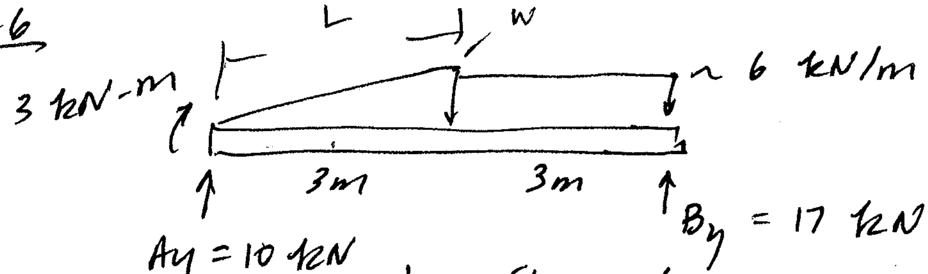
$$17x = 3 - x \qquad 18x = 3$$

$$x = .1667$$

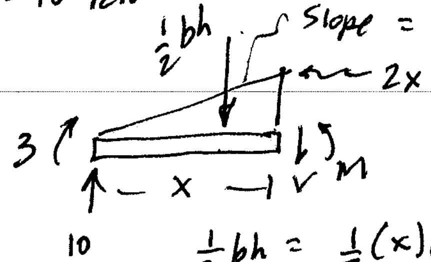
$$3 - x = 2.833$$

$$A_2 = \tfrac{1}{2}(.1667)(1) = .0833$$

F7-6



$0 \le x \le 3$



$\left(\frac{w}{L}x\right)$

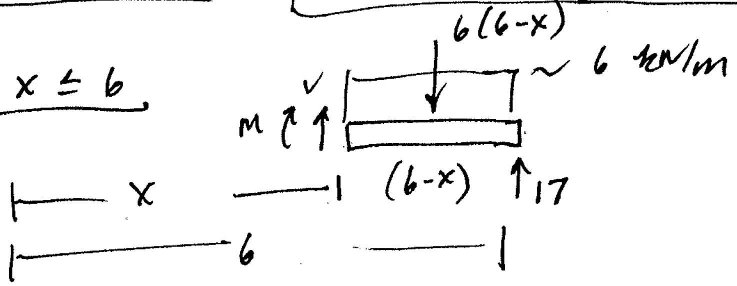
$\frac{1}{2}bh = \frac{1}{2}(x)(2x) = x^2$

$+\uparrow \quad 10 - x^2 - V = 0; \qquad \boxed{V = 10 - x^2}$

$+\circlearrowleft \Sigma M_{cut} = 0; \qquad M - 10(x) - 3 + x^2\left(\frac{x}{3}\right) = 0$

$$\boxed{M = 10x + 3 - x^3/3}$$

$3 \le x \le 6$



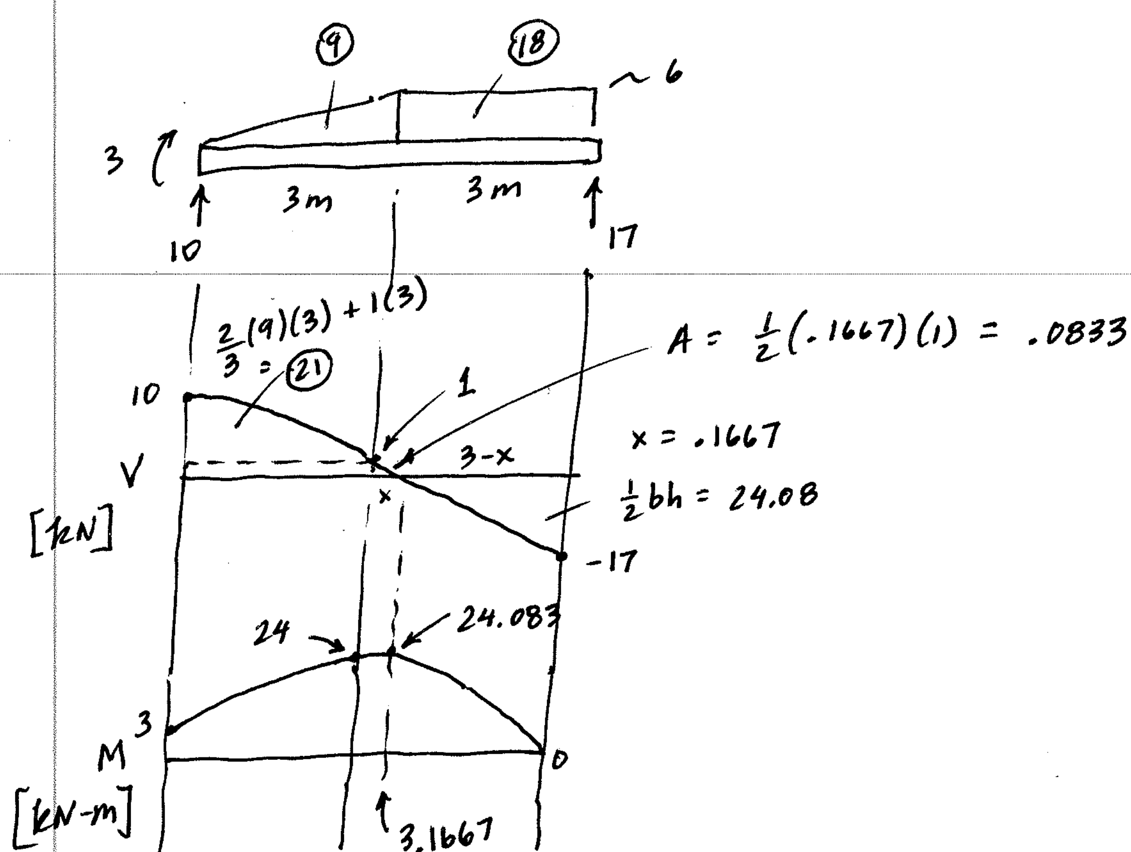
$+\uparrow \quad V + 17 - 6(6-x) = 0$

$$\boxed{V = 6(6-x) - 17}$$

$+\circlearrowleft \Sigma M_{cut} = 0; \qquad M - 17(6-x) + 6\frac{(6-x)^2}{2} = 0$

$$\boxed{M = 17(6-x) - 3(6-x)^2}$$

F7-6 (cont'd)



Test Eqns

$0 \le x \le 3$

$V = 10 - x^2$

$M = 10x + 3 - \frac{1}{3}x^3$

Note: $M = \int V\,dx + C \ldots$

x	V	M
0	10	3
3	1	24

checks!

$3 \le x \le 6$

$V = 6(6-x) - 17$

$M = 17(6-x) - 3(6-x)^2$

$M = \int V\,du$ where $u = (6-x)$

x	V	M
3	1	24
3.1667	0	24.084
6	-17	0

checks!